

EDUCATION

Columbia University

M.S. in Computer Science

Dean's MS Academic Excellence Fellowship - \$15,000

New York, NY

Expected Dec. 2027

University of Massachusetts Lowell

B.S./M.S. in Computer Science; GPA: 4.0/4.0

Chancellor's Medals for Distinguished Academic Achievement

Lowell, MA

Jan. 2024 – Dec. 2025

SKILLS

- **Programming Languages:** Java (4 yrs), Javascript/Typescript (4 yrs), Python (3 yrs), Go, C++, Kotlin, PHP, Lua, Rust, Bash, Markdown, SQL, HTML/CSS, LaTeX, XML, TOML
- **Frameworks and Libraries:** Spring Boot, Node.js, React, MUI, NestJS, Vite, JUnit, Nest.js, Drizzle ORM, Numpy, Pandas, Matplotlib, Scikit-learn, PyTorch, Typer
- **Databases and APIs:** MySQL, PostgreSQL, MongoDB, Redis, HTTPS, RESTful, GraphQL, WebSockets, Swagger
- **Tools and Platforms:** Git, GitHub, Nginx, systemd, Docker, AWS EC2/Lightsail, POSIX, npm/pnpm/bun, Neovim, Postman
- **AI:** OpenAI API, Kaggle, NLP, Neural Networks, Transformer/GPT, LoRA, RLHF, PPO

WORK EXPERIENCE

UMass Lowell - Miner School of Computer and Information Sciences

Software Engineering Intern

Lowell, MA

May. 2025 – Dec. 2025

- Migrated 28k+ AEPW (Alliance to End Plastic Waste) records into a analytical PostgreSQL schema, cutting query times by about 70% and improving data integrity.
- Designed and implemented AUDA, a Python analytics platform for data cleaning, preprocessing, model execution, and visualization.
- Built and evaluated machine learning workflows for sparse country-level time series, including Isolation Forest anomaly detection, Random Forest feature importance, and forecasting/reconstruction with automated hyperparameter optimization.
- Developed a Typer-based CLI to automate analytics workflows, improving reproducibility, maintainability, and developer productivity.

Take 2 Health

Backend Developer Intern

Shenzhen, China

Mar. 2021 – Sep. 2021

- Engineered the backend for a blood specimen management platform using Midway.js, implementing JWT-based authentication and RESTful APIs for user and specimen operations.
- Created Swagger API documentation to streamline integration and testing, and collaborated with frontend engineers to ensure reliable interface behavior.
- Architected a microservice-based system using gRPC for high-performance inter-service communication, improving scalability and modularity.

Juyingtong

Backend Developer

Zhuhai, China

Nov. 2018 – Feb. 2020

- Developed the backend for a WeChat mini-program for conference management using ThinkPHP 5 and MySQL, with Redis-backed session authentication and RESTful APIs for user and event operations.
- Built a location-aware attendee check-in system using QR code scanning, encrypted conference metadata, WeChat geolocation validation, and real-time attendance recording.
- Integrated the WeChat Pay API to support secure, real-time payment processing for conference registration and related services.
- Deployed the application on Alibaba Cloud Linux servers with Nginx reverse proxy and load balancing, using systemd for process management.

PROJECTS

- **Burrow:** An extensible command-line application platform that lets users create independent CLI applications (chambers) composed of extensions, which can start in under 20 milliseconds. There are built-in modules for chamber management, configuration editing, HTTP server startup, and self-build/update workflows.
- **BG Clock:** A responsive touchscreen clock application for board games like Go and Chess, built with **Vite**, **React**, and **Chakra UI**. It has an internal modular game core with a precise game loop for accurate timekeeping, and an extensible trigger system for flexible mode support. (Live Demo: <https://bgc.jameschen.life>)
- **(Snowy) Outpost:** A tool that creates a temporary high-performance DigitalOcean droplet (a cloud instance) and hosts a Minecraft server there. It automatically uploads the game directory on start and downloads it on stop.